

CS336 Spring 2025 Assignment 2: Systems Writeup

Contents

1 Problem: benchmarking_script	1
2 Problem: nsys_profile	2
3 Problem: mixed_precision_accumulation	2
4 Problem: benchmarking_mixed_precision	2
5 Problem: memory_profiling	2
6 Problem: pytorch_attention	2
7 Problem: torch_compile	2
8 Problem: flash_forward	2
9 Problem: flash_backward	2
10 Problem: flash_benchmarking	2
11 Problem: distributed_communication_single_node	2
12 Problem: naive_ddp	3
13 Problem: naive_ddp_benchmarking	3
14 Problem: minimal_ddp_flat_benchmarking	3
15 Problem: ddp_overlap_individual_parameters	3
16 Problem: ddp_overlap_individual_parameters_benchmarking	3
17 Problem: ddp_overlap_bucketed	3
18 Problem: ddp_bucketed_benchmarking	3
19 Problem: communication_accounting	3
20 Problem: optimizer_state_sharding	3

1 Problem: benchmarking_script

(a) Deliverable

Implemented in `cs336_systems/benchmark.py`. The script initializes the model from CLI hyperparameters, creates random input data, runs warmup steps, then measures forward-only or forward+backward timing with `timeit.default_timer()` and `torch.cuda.synchronize()`.

(b) Forward/backward timings (5 warmup, 10 measured steps)

Model	Fwd Mean (ms)	Fwd Std (ms)	Bwd Mean (ms)	Bwd Std (ms)	Step Mean (ms)	Step Std (ms)
small	105.303	38.286	62.392	30.468	170.464	38.24
medium	184.545	38.394	85.389	23.722	288.648	25.34
large	339.047	2.485	156.582	1.572	499.472	3.68
xl	733.624	42.155	313.152	1.035	1099.266	5.91
2.7b	838.264	9.618	457.806	1.187	1307.808	24.51

Table 1: Part (b) benchmark results on CUDA (A100 40GB).

Forward and backward time increase with model size. Variability is generally low for larger models in this run (especially backward), while small/medium show higher standard deviation.

(c) Effect of fewer/no warmup steps

Model	Warmup 0		Warmup 1		Warmup 2		Warmup 5	
	Mean	Std	Mean	Std	Mean	Std	Mean	Std
small	225.844	138.148	169.072	36.043	176.863	30.199	170.464	38.242
medium	344.245	181.775	294.368	34.751	295.297	25.260	288.648	25.349
large	596.205	239.325	570.078	35.475	518.251	32.955	499.472	3.688
xl	1186.364	289.365	1077.503	31.389	1092.626	12.791	1099.266	5.916
2.7b	1402.911	225.944	1321.353	34.838	1319.389	37.589	1307.808	24.517

Table 2: Part (c): step total timing (ms), 10 measured steps, varying warmup count.

Without warmup ($w=0$), both average step time and variance are much worse across sizes. Using 1–2 warmup steps substantially improves stability, but can still differ from the 5-step baseline because one-time startup effects (kernel autotuning/selection, allocator behavior, cache setup, lazy initialization) may not be fully amortized yet.

2 Problem: nsys_profile

TODO: Add results and responses for parts (a)–(e).

3 Problem: mixed_precision_accumulation

TODO: Add mixed-precision accumulation findings.

4 Problem: benchmarking_mixed_precision

TODO: Add timings and commentary for full vs. mixed precision.

5 Problem: memory_profiling

TODO: Add memory timeline images, peak-memory table, and responses for parts (a)–(e).

6 Problem: pytorch_attention

TODO: Add benchmark table, memory accounting, and discussion.

7 Problem: torch_compile

TODO: Add compilation benchmarking results and response.

8 Problem: flash_forward

TODO: Add implementation notes and requested deliverables.

9 Problem: flash_backward

TODO: Add backward implementation results and validation.

10 Problem: flash_benchmarking

TODO: Add FlashAttention benchmarking tables and analysis.

11 Problem: distributed_communication_single_node

TODO: Add distributed communication measurements and response.

12 Problem: naive_ddp

TODO: Add naive DDP implementation deliverable details.

13 Problem: naive_ddp_benchmarking

TODO: Add naive DDP benchmarking results and discussion.

14 Problem: minimal_ddp_flat_benchmarking

TODO: Add flat-buffer DDP benchmarking results.

15 Problem: ddp_overlap_individual_parameters

TODO: Add implementation and correctness evidence.

16 Problem: ddp_overlap_individual_parameters_benchmarking

TODO: Add overlap benchmarking results and conclusions.

17 Problem: ddp_overlap_bucketed

TODO: Add bucketed overlap implementation details.

18 Problem: ddp_bucketed_benchmarking

TODO: Add bucket-size benchmarking results and analysis.

19 Problem: communication_accounting

TODO: Add communication-volume accounting calculations.

20 Problem: optimizer_state_sharding

TODO: Add implementation details and validation.

21 Problem: optimizer_state_sharding_accounting

TODO: Add memory/speed accounting and ZeRO comparison.